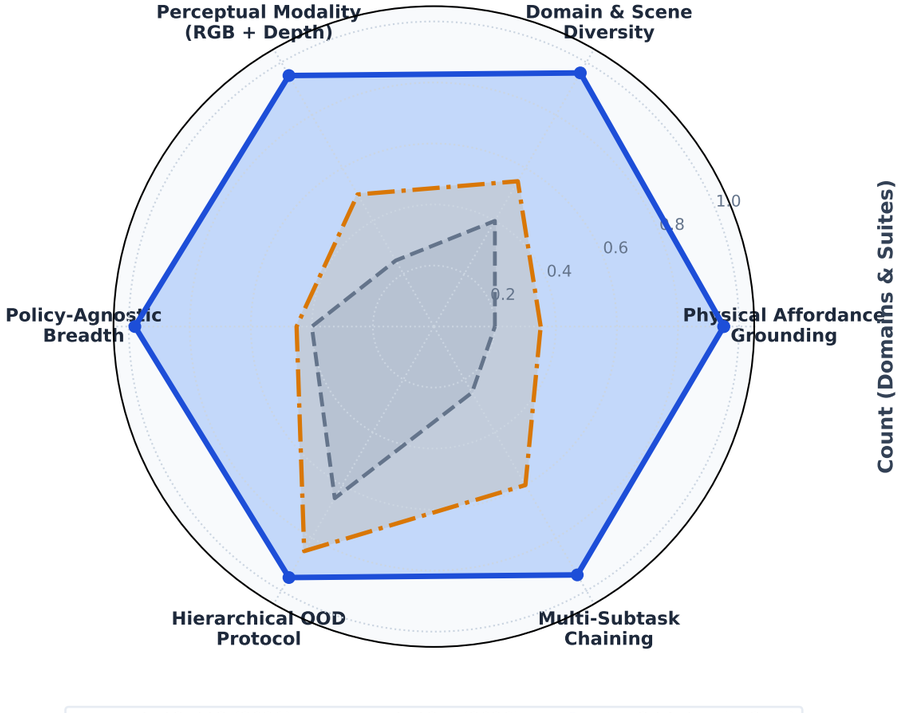
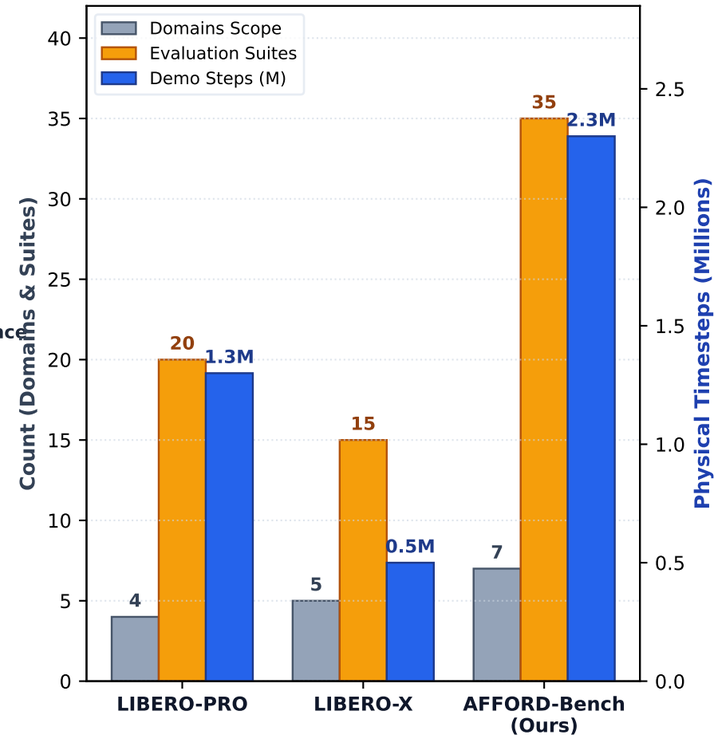
AFFORD-Bench vs. LIBERO-PRO & LIBERO-X: Architecture, Scale, and Physical Grounding

(a) Multi-Dimensional Capability Radar



(b) Quantitative Scale Comparison



(c) Architectural & Physical Grounding Comparison

Benchmark Feature	LIBERO-PRO	LIBERO-X	AFFORD-Bench (Ours)
Friction & Payload	Nominal Only	Nominal Only	**μ ∈ [0.1, 1.2], 0.5-3× m**
Insertion Clearance	Coarse Tabletop	Coarse Tabletop	**Sub-cm (±1.5–5 mm)**
Sensory Streams	1× RGB (128x128)	**2× RGB (256x256)**	**Dual RGB + 3D Depth**
Compound Prompts	Atomic Single-Step	**Subtask Tags**	**Chained 2–3 Step Continuous**
Policy Agnosticism	VLA Models	VLA Models	**BC, DP, ACT, 3D, VLA, RL**
Demonstration Data	6,500 Demos	2,520 Demos	**10,500 Demos (2.3M Steps)**
Simulation Engine	Robosuite (~15 FPS)	Robosuite (~15 FPS)	**Native MuJoCo (45 FPS, $0)**